

Agenda :-

1) Intro to LLD

2) Why LLD is important?

- * Day to Day job

- * Interviews.

3) Structure of LLD Module

4) Naunces / Expectation Setting

1) why language (java) ✓

2) Assignments ✓

3) amount of theory

4) CS fundamentals

5) Job Readiness.

What is HLD?

H : how

L : level

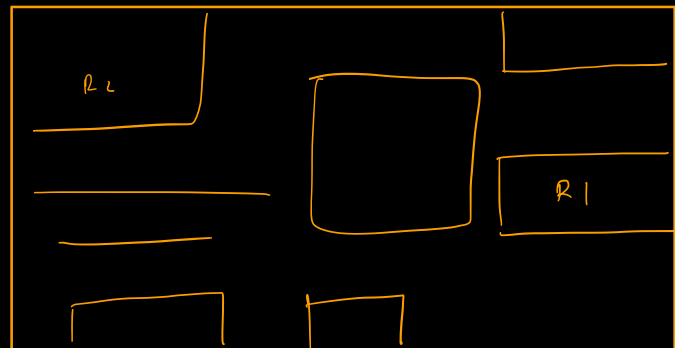
D : Design

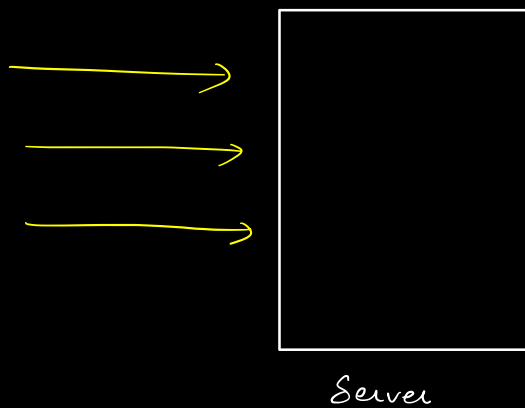


Overview /
brief (didn't
go through
detail) /
Superficial.

PM → PD → SWE

Before we create anything, we document how
that will look like.





i) Huge waiting time for req (slow)

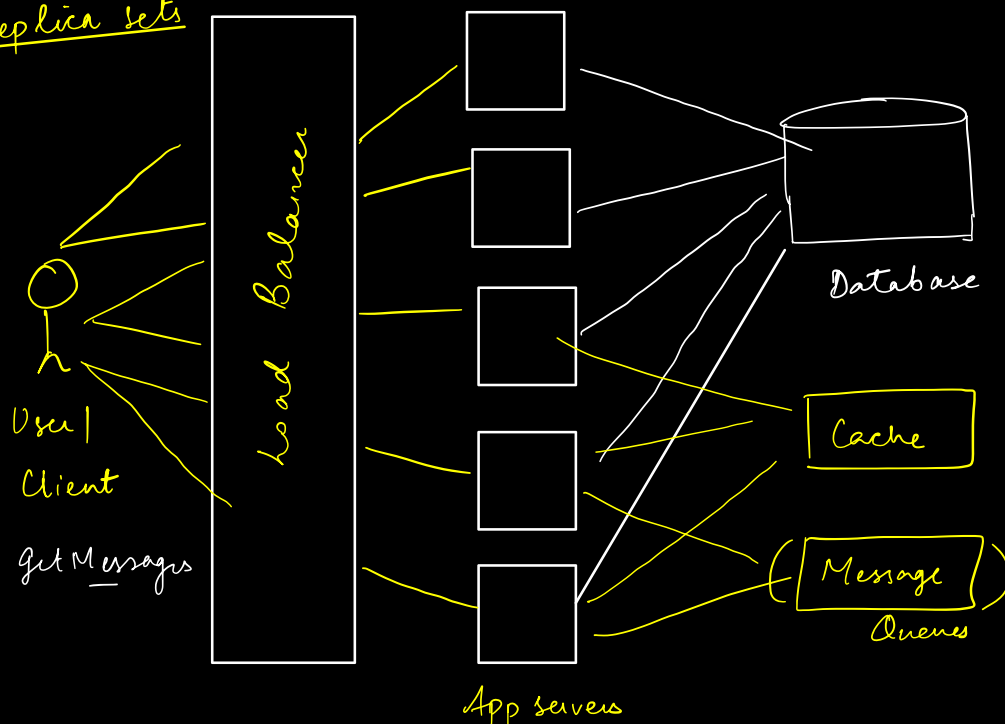
ii) Breakdown / crash

iii) Single point of failure

iv) User experience will be bad

Geo DNS

Replica sets



Overview of different infra layers that work together to serve an application at scale &

described efficiency.

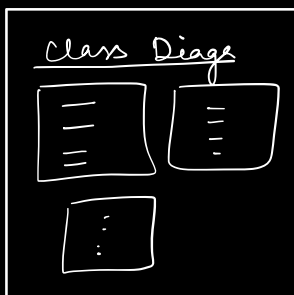
At the end, each component is nothing but a computer which acts as per what program is running on that computer.

Detail $\rightarrow$ Detail of what ^{code of} program is running.

LLD :- Design of code that you are going to write.

Details of SW

Implementation of the SW system.



Details of
 $\rightarrow$ Code

l1d

✓ HKD : Architects | SDE 3+

✓ LLD : SDE 2+ | Implement

$\rightarrow$ SD

$\rightarrow$ Design SW

LLD : OOD Design

Why LHD is important!

i) Day to Day job?

On an avg working day how many hr you write
code $\Rightarrow$

12% $\Rightarrow$ writing code.

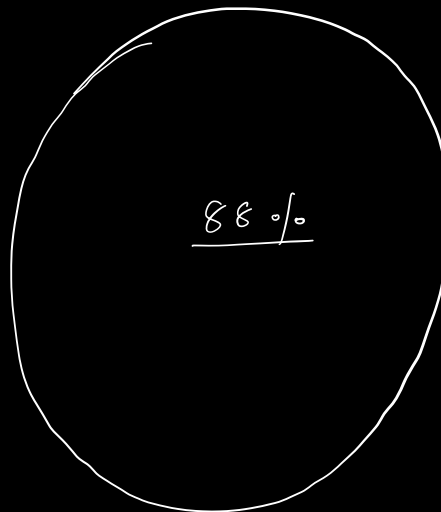
Documentation (Design)

Testing

Debugging / Optimization

Meeting

Code Review



Design

Understand the code.

Req Gathering

Testing

$\hookrightarrow$ 88% of the time in a better way

↑
LHD

LHD is used to write code that is :-

Understandable

Reusable

Extensible

Maintainable

Modular.

Extensibility vs Maintainability :-

Extensive :- (New Req)

how easy it is for your \$lw system
to be modified to be able handle new
requirements.

Maintainable :- (No new Req)

- bugs
- dependency updates .
- perf optimization
- platform updates .

Interviews ?

* Every dream startup is going to have atleast
1 LLD round in the interview.
(even if you are a fresher)

* Most startups have first round itself as the
lld round.

Leetcode EM

<u>exp</u>	
0-2	SDE-1
2-5	SDE-2
5+	SDE-3+

DSA is
common...

Types of → Companies	Startup (Unacademy, Cred, Paytm, Flipkart, Sharechat, Razorpay)	MNC (Oracle, Adobe, Visa, Paypal, Cisco)	FAANGM FB } <u>No LLD</u> Google }
SDE 1	Machine Coding Round	NONE ✓ <u>Theoretical Ques</u>	<u>LLD</u> OOPs Design principles
SDE 2	"	LLD (1hr)	LLD ✓
SDE 3+	"	LLD (1hr)	LLD- ✓

3 hr interview

Machine Coding

Design

* Class Design ✓

* Schema Design

+

Working Code - X

Structure of LLD Module :-

LLD 1:- Foundational Design / Language concepts.

Whatever theory / foundation are important for

- OOP (4 classes) ① Design
- Concurrency / Multithreading (4 classes) ② Interviews Ques.
- Adv Language concepts (4 classes)

Java Streams

Lambda fun

Collections.

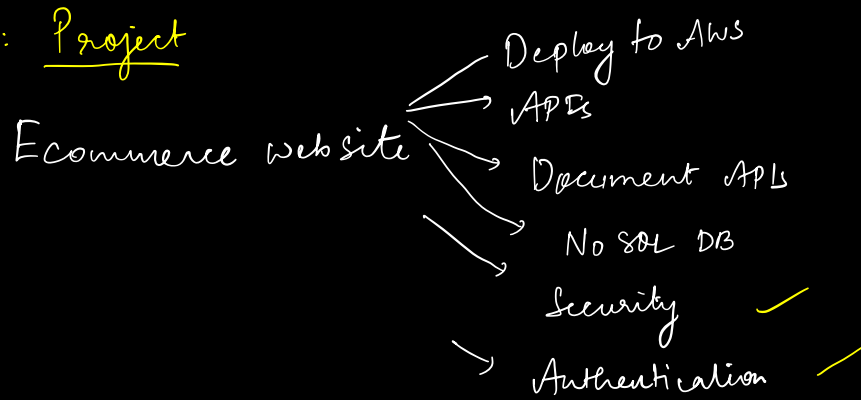
LLD 2:- How to do better design?

- SOLID design principles? (2 classes)
- Design pattern (9-10 class) } minor
- UML Diagram (1 class) } Case Studies

LLD 3:- Machine Coding Case Studies - (Mini project)

- Pen
 - Tic Tac Toe
 - Parking Lot
 - Book my show
 - Splitwise
 - Cache
- MVC
Microservices
Repository DP
Intro to spring boot
Work with DB

LLD 4 : Project



Java Py Rest
85 : 10 : 5

CS fundamentals:-

DBMS

↳ SQL classes

OS

↳ LLD-1 Module

↳ CPU Scheduling

↳ Memory manag

CN

LLD-4 cover it. ✓

Job Readiness

SDE-1

Apply when

DSA

+ Language

+ SOL + LLD

(lld.3)

is complete

SDE 2+

DSA

+ Lang

+ LLD

+ SOL

+ HLD

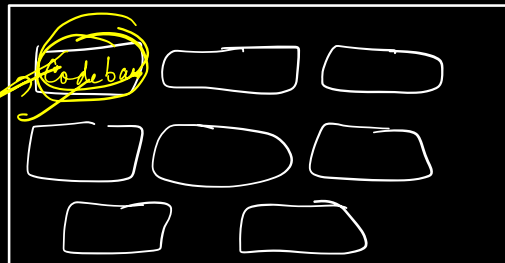
is complete

Please Grit

learn about

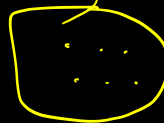
✓✓✓
VCS

foldo
files



write

Compan



Crit



On



Crit